



## Original Research Article

# Promotion of cellular differentiation and DNA repair potential in brain cancer cells by Shankhpushpi, (*Clitoria ternatea* L.) with rasayana properties *in vitro*

Keshava Prasad<sup>a,1</sup>, Kothanahalli S. Raghu<sup>b,1</sup>, Swathi Maruthiyodan<sup>b</sup>, Renu Wadhwa<sup>c</sup>,  
Sunil C. Kaul<sup>c</sup>, Kapaettu Satyamoorthy<sup>a,d</sup>, Kanive Parashiva Guruprasad<sup>e,\*</sup>

<sup>a</sup> Department of Cell and Molecular Biology, Manipal School of Life Sciences, Manipal Academy of Higher Education, Planetarium Complex, Manipal, 576 104, Karnataka, India

<sup>b</sup> Department of Ageing Research, Manipal School of Life Sciences, Manipal Academy of Higher Education, Planetarium Complex, Manipal, 576 104, Karnataka, India

<sup>c</sup> Drug Discovery and Assets Innovation DBT-AIST International Laboratory for Advanced Biomedicine (DAILAB), Biomedical Research Institute, National Institute of Advanced Industrial Science & Technology (AIST), Tsukuba, 305 8565, Japan

<sup>d</sup> Shri Dharmasthala Manjunatheshwara (SDM) University, SDM College of Medical Sciences and Hospital, Manjushree Nagar, Sattur, Dharwad, 580009, Karnataka, India

<sup>e</sup> Centre for Ayurvedic Biology, Department of Ageing Research, Manipal School of Life Sciences, Manipal Academy of Higher Education, Manipal, 576104, Karnataka, India



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## ABSTRACT

**Background:** Brain ageing is accompanied by the diminution of neuronal plasticity, which is correlated with the inability to respond to loss of memory, various stress-induced stimuli, and increased risk of neurodegenerative disorders. In the recent past, plant based herbal medicines are of interest over synthetic drugs for therapeutic purposes due to lower side effects. The Indian traditional medicine Ayurveda describes several herbal remedies, such as rasayana (elixirs for rejuvenation), to treat many age-related diseases. Medhya rasayana (brain elixir) is a class of rasayana used for its nootropic functions, such as enhancement of memory and intellect, in addition to promoting normal health. *Clitoria ternatea* L. is one such plant used in the preparation of medhya rasayana.

**Objective:** To investigate the neuronal differentiation and DNA repair potential of Shankhpushpi (*Clitoria ternatea* L.) in neuroblastoma cells.

**Materials & methods:** The effect of *Clitoria ternatea* L. on neuronal cell differentiation, DNA repair (base excision repair, comet, γH2AX immunostaining assays), autophagy by cadaverine uptake and mitochondrial functions by fluorescent dye staining through flow cytometry were evaluated.

**Results:** The results revealed that *Clitoria ternatea* L. enhanced DNA repair and mitochondrial membrane potential and reduced autophagy and reactive oxygen species (ROS) in IMR-32 neuroblastoma cells. Treatment of IMR-32 neuroblastoma and C6 glioblastoma cells with shankhpushpi induced neuronal differentiation and exhibited markers such as MAP2, Mortalin and GFAP.

**Conclusion:** Neurobiological pathways and molecular mechanisms influenced by rasayana herb shankhpushpi suggests its therapeutic potential for neurodegenerative diseases.

## 1. Introduction

Traditional systems of medicine native to the Indian subcontinent, such as Ayurveda, and its practices include the use of herbs, minerals,

metal supplementations and surgical procedures [1]. Rasayana, prepared traditionally from a mixture of defined herbs, has been used extensively to promote healthy ageing [2]. Being one among the eight branches of Ayurveda, the *Rasayana tantra* focuses on use of herbal

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\* Corresponding author. Department of Ageing Research Manipal School of Life Sciences Manipal Academy of Higher Education Manipal, 576 104, Karnataka, India.

E-mail address: [guruprasad.kp@manipal.edu](mailto:guruprasad.kp@manipal.edu) (K.P. Guruprasad).

<sup>1</sup> These authors equally contributed to this work.

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tonics in tissue rejuvenation and repair through nutrient supplementation, metabolic alterations and tissue nourishment [3,4]. These herbal formulations are reported to improve immunity, and physical and mental health and may also regulate the homeostasis of cells and tissues [5]. Puri, (2003) described Rasayana as those herbal formulations which on proper administration and use, delay ageing, improve intellect, vital energy and eyesight and aid in the overall rejuvenation of the body and strengthen the primordial tissue rasa [2]. The status of Dhatus (tissue) is thus promoted through the physio-chemical action encompassed by rasayana [3]. Rasayana can also be compared to alternatives. Alternatives work as blood cleansers and strengthen the functions of the body by promoting health and vitality. However, rasayana is traditionally administered under strict dietary and environmental control. At the cellular level, telomere length shortening, retarded telomerase activity and DNA damage by exogenous or endogenous stress contribute to ageing [6]. The rasayana formulations act on DNA repair pathways, boosting them, reducing clastogenicity and enhancing cytoprotective effects thereby suppressing age-related disorders mechanistically [6–9]. Despite its claim to healthy ageing, the underlying cellular and molecular mechanisms and functions in repairing, rejuvenating and enhancing the quality of the cells remain to be fully understood. As Ayurveda does not describe the molecular mechanisms involved with rasayana effect, it is apparent to investigate these aspects through modern science. Thus, help connect Ayurveda with modern medicine.

‘Medhya rasayana’ is a nootropic rasayana formulation that is frequently used to improve mental ability, intellect, cognition and promote brain functions [5]. Medhya rasayana helps in delaying brain ageing, improving neural tissue generation by reducing stress and inducing adaptogenic effects [3]. Shankhpushpi is one of many plants possessing medhya (nootropic) rasayana properties [10–16]. The texts of Ayurveda such as Charaka Samhita (Chikitsa Sthana, Rasayana Chapter, Karaprachitiyo rasayanapada, Medhya rasāyanāni, Verse 30–31) have also described the use of shankhpushpi in medhya rasayana preparation. Although many plant species are identified as shankhpushpi by traditional practitioners [13,14], *Clitoria ternatea* L. (CT), a perennial climber that belongs to the family Fabaceae, sub family Papilionaceae is considered as shankhpushpi and is extensively used as medhya rasayana plant in South India [13,14,17,18]. It is also referred as Aparajita, Gokarna, and Vishnukranta [18]. The extracts of this plant have a wide range of secondary metabolites which are the key to its healing abilities in the treatment of various neurodegenerative disorders [8,10,17,19]. A list of phytochemical constituents of *Clitoria ternatea* L. includes kaempferol, quercetin, myricetin, ternatins, delphinidin, cyclotides and many others [20]. The roots of the plants are used to treat dementia and burning sensation, while the leaves are used in otalgia and hepatopathy [21]. The plant possesses neurogenic potential [22], anti-cervulansant, anti-stress [23] and anti-depressant activities [24]. Enhancement of learning and memory is also observed by *Clitoria ternatea* L. extract in rodents [14,22]. Furthermore, an increased proliferation rate along with growth of neurospheres and differentiation of neural stem cells (NSCs) of the anterior subventricular zone has been reported [25]. It was further demonstrated that the subtoxic aqueous root extract of *C. ternatea* has a significant effect on various metabolic processes of neuronal cells, such as the survival, proliferation, differentiation and growth of newborn neuronal cells [26]. Rai and coworkers (2005) have shown that treatment with *Clitoria ternatea* L. increases acetylcholine levels in the brain region and enhances the development of neurons in the amygdala region, the memory center of the brain (hippocampus), suggesting improvement of neuronal functions [25]. Additionally, *Clitoria ternatea* L. induces neurogenesis in the amygdala, which might enhance therapeutic potential in amygdala-related neurodegenerative disorders [10,25]. A study on diabetic rat models showed marked increase in superoxide dismutase (SOD), glutathione (GSH) and catalase (CAT) levels [10,27]. Although these reports suggest the effect of *Clitoria ternatea* L. on neuronal functions, the underlying cellular and molecular mechanisms of action are not clearly understood, hence,

deciphering them may aid in potential therapeutic applications. Thus, the study focuses on understanding the effects of *Clitoria ternatea* L. (shankhpushpi) on DNA damage and repair, autophagy, mitochondrial functions and the expression of neuronal differentiation markers in human neuroblastoma IMR-32 and rat glioblastoma C6 cells *in vitro*, which are implicated in neuronal functions and ageing.

## 2. Materials and methods

### 2.1. Preparation of shankhpushpi extracts

Shankhpushpi plants were grown at the herbal garden of Manipal School of Life Sciences, MAHE, Manipal, and were used for the preparation of medhya rasayana like formulations according to Ayurvedic literature [28] as per the traditional use by Ayurveda practitioners [8]. The aqueous extract of this rasayana like formulation was prepared for the present *in vitro* study. In brief, the freshly harvested whole plant and jaggery (1:1) were blended to mix properly. 20 g of this rasayana like formulation was dissolved in 100 mL Milli-Q water and incubated overnight at 45 °C with shaking at 100 rpm. This was filtered, sterilized and considered as 100% extract. The same procedure without using jaggery was also carried out to prepare the aqueous extract of *Clitoria ternatea* L. for the analysis of differentiation markers. The preparations were stored at –80 °C until further use.

### 2.2. Phytochemical profiling by Mass spectrometry

Phytochemical profiling was performed in mass spectrometry to determine the constituent phytochemicals present in shankhpushpi plant aqueous extracts [8]. The extract was lyophilized and reconstituted in 0.1% formic acid in methanol. This reconstituted sample (5 µL, 4 µg/µL) was then run in a reverse phase HPLC-mass spectrophotometry using C-18 column (1200 series/6520-Accurate Mass Q-TOF-LC/MS; Agilent MassHunter Qualitative-Analysis software version B.04.01; Agilent technologies). 0.1% formic acid in water was used as mobile phase A, whereas 0.1% formic acid in 90% acetonitrile was used as mobile phase B with a flow rate of 400 µL/min. Elution was completed by a gradient increase of mobile phase B (5–35% B from 0 to 35 min followed by 35–75% B in 35–45 min). The phytochemicals were identified by confirming the *m/z* values from plant metabolite databases [29,30].

### 2.3. MTT cytotoxicity assay

An equal number of IMR-32 cells ( $5 \times 10^3$  cells/well) in 96-well plates were incubated with shankhpushpi extract (0.5%–6%) for 24 h, 48 h and 72 h with appropriate controls. The percentage viability was calculated based on reduction in tetrazolium salt (MTT) carried out in living cells to form a blue formazan measured spectrophotometrically at 570 nm.

### 2.4. Cell culture and treatment

The IMR-32 human neuroblastoma and C6 rat glioblastoma cells were cultured using DMEM in 10% FBS at 37 °C in a 5% CO<sub>2</sub> incubator (Hera Cell-150, Thermo Scientific). At 60–70% confluency, they were treated with shankhpushpi extract for 24 h. These cells were used for various assays including DNA repair assays.

### 2.5. DNA repair assays

#### 2.5.1. Comet assay

The assay was performed as mentioned in Olive and Banath [31] with few modifications. IMR-32 cells treated with and without shankhpushpi extract (2%) were subjected to UV irradiation (1J/cm<sup>2</sup>) and incubated for 0 h, 1 h and 2 h to test the efficacy of damaged DNA

repair. The frosted microscopic slides were pre-coated with agarose by dipping them in 1.5% molten agarose (made up in filtered PBS) and the slides were dried. Cells (~2000 cells) from each sample in 20  $\mu$ L of PBS, were mixed with 180  $\mu$ L 0.75% agarose (low melting) and pipetted immediately on to a pre-coated slide and sandwiched by coverslip (22  $\times$  22 mm). After placing the slides on ice for 2 min (as soon as agarose solidified), coverslips were removed by sliding. Finally, 200  $\mu$ L of 0.75% agarose (low melting) was poured on the slide and covered with cover glass. The slides were placed on ice again for 2 min to allow agarose to form a gel, and the cover slips were removed by sliding. The slides were then covered in staining trays with ice cold lysis solution (2.5 mM NaCl, 100 mM EDTA, 10 mM Tris, in 1% Triton X 100 and 10% DMSO, pH 10) and incubated for overnight at 4  $^{\circ}$ C. The slides were covered with alkaline electrophoresis solution (NaOH, EDTA, pH 13, 4  $^{\circ}$ C) for 20 min. Subsequently, slides were placed flat onto a gel tray in a horizontal electrophoresis system aligned equidistant from the electrodes. A voltage of 20 V and current of 300 mA were set and electrophoresed for 25 min. Thereafter, few drops of neutralising buffer (0.4 M Tris HCl, pH 7.4) were gently poured on the slides and stained with ethidium bromide (150  $\mu$ L–10 mg/mL). For visualization, a fluorescence microscope with a 515–535 nm excitation filter and 590 nm barrier filter at 10x magnification was used. The Olive tail moment (OTM) was analysed in approximately 55–60 cells per sample using the Komet package (Version. 5.5, Kinetic Imaging Ltd., Bromborough, UK). The mean OTM is calculated as (Head Mean)  $\times$  Tail % DNA/100.

#### 2.5.2. Base excision gap repair/BER assay

The harvested cells were washed in ice-cold PBS. Protein was extracted using cell lysis buffer (40 mM HEPES, 500 mM NaCl, 10% Glycerol, 0.1% NP-40, pH 7.5) containing protease inhibitor cocktail (Thermo Fisher), and its concentration was estimated by Bradford's method. The BER assay was performed as described in Raghu et al. (2017) [8]. Briefly, a 20  $\mu$ L reaction mixture containing 5  $\mu$ g protein from whole cell extract in 200 mM HEPES-NaOH (pH 7.5), 1 mM DTT, 10 mM MgCl<sub>2</sub>, 0.1 mg/mL BSA, 20% glycerol, 200  $\mu$ M each of the dNTPs and annealed oligomer duplex with one nucleotide gap (30 ng/ $\mu$ L) was incubated for 90 min at 37  $^{\circ}$ C. Subsequently, to stop the reaction, 20  $\mu$ L formamide was added and kept on boiling water bath (95  $^{\circ}$ C) for 10 min. Thereafter, the samples were instantly snap-cooled, run on a urea polyacrylamide (20%) denaturing gel at 250 V, and fluorescence was imaged at 520 nm in an Amersham Imager 680 (GE Healthcare). The repair capacity was estimated by quantifying the repaired oligonucleotide (complete repair) with the total oligonucleotide (repaired and unrepaired). This experiment was repeated three times, and the mean values of three independent experiments were used.

#### 2.5.3. Cell cycle and apoptosis analysis by flow cytometry

Approximately  $5 \times 10^5$  IMR-32 cells/6 cm petri plates were incubated with shankpushpi extract (2%) for 24 h. Subsequently, the harvested cells were fixed in 70% ethanol with gentle mixing and incubated at 4  $^{\circ}$ C for 30 min. The cells were then incubated in 15  $\mu$ g of RNase (Sigma) and 0.1% Triton X-100 for 30 min at 37  $^{\circ}$ C and centrifuged at 3000 rpm for 10 min. This was followed by resuspension of the cells in 0.5 mL of PBS with 10  $\mu$ g propidium iodide with 1 h incubation in the dark. Later, the samples were analysed by flow cytometry (BD Biosciences, San Jose, USA). For calculation of the percentage of apoptotic cells, the ratio of the fluorescent area (sub-G1) below the G0/G1 peak value to the total area of fluorescence was used.

#### 2.5.4. DPPH and ABTS free radical scavenging activity

The DPPH assay was carried out using serial concentrations (2–24%) of shankpushpi aqueous extract by following the method as described [15,32]. In brief, the extracts were placed in 96-well plate and made up to 100  $\mu$ L with Milli-Q water. DPPH (150  $\mu$ L; 200  $\mu$ g/ml in absolute methanol) was added to each well and the absorbance was measured at 517 nm. In the ABTS assay, different concentrations (0.6–3.6%) of

shankpushpi extract were used in 96-well plates and brought up to 10  $\mu$ L with Milli-Q water. To each well, 240  $\mu$ L of ABTS working solution was added, and the absorbance of 740 nm was measured using a Tecan Infinite M200 microplate reader (Tecan, Austria) [32]. Ascorbic acid was used as a standard.

#### 2.5.5. Mitochondrial assays - mitochondrial mass (MM), mitochondrial membrane potential (MMP) and ROS

Approximately  $3\text{--}5 \times 10^5$  IMR-32 cells/6 cm petri plates were incubated with shankpushpi aqueous extract (2%) for 24 h. The medium was discarded, and the cells were washed twice with PBS, and incubated at 37  $^{\circ}$ C for 30 min with serum-free medium containing 5  $\mu$ M DCFDA for ROS, Rhodamine-123 for MMP or NAO for MM. Cells were collected and resuspended in 500  $\mu$ L of PBS, and ROS, MMP and MM were then analysed by flow cytometry (Becton Dickinson FACS Calibur).

#### 2.5.6. Phospho- $\gamma$ H2AX immunostaining

IMR-32 cells were grown on 6-well plates at a seeding density of  $5 \times 10^5$  cells. The cells were treated with shankpushpi aqueous extract (1% and 2%), or shankpushpi aqueous extract (1% and 2%) + 50 nM doxorubicin for 24 h or 48 h. After treatment, the cells were trypsinized and centrifuged, and the cell pellet was collected in microfuge tubes. The pellet was fixed in 4% paraformaldehyde for 15 min with gentle vortexing to avoid cell aggregation. The cells were washed in PBS (pH 7.4), and centrifuged, and the cell pellet was collected. This was followed by blocking in 0.1% Triton X containing 5% BSA for 1 h. Later, the cells were incubated in primary rabbit anti-phospho- $\gamma$ H2AX IgG (1:1000) (catalog no: NB100-384) from Novus Biologicals, overnight at 4  $^{\circ}$ C. The cells were then washed in PBS (pH 7.4) and incubated with secondary FITC AffiniPure Alpaca anti-rabbit IgG antibody (1:300) (catalog no: 615-095-215) from Jackson ImmunoResearch USA for 1 h in the dark at 4  $^{\circ}$ C. The cells were later incubated with Hoechst (1:500) for 15 min and analysed in CyFlow Space (Partec/Sysmex India Pvt. Ltd) flow cytometer using FloMax software version 2.7.

#### 2.5.7. Immunofluorescence staining

IMR32 and C6 cells grown on coverslips were fixed using paraformaldehyde and then permeabilized (PBS with 0.3% Triton X-100). The IMR 32 cells were then incubated with anti-growth associated protein 43 (GAP43), anti-neurofilament Heavy (NF-H), anti-post synaptic density (PSD) or anti-microtubule associated protein (MAP2) antibodies diluted in 0.1% PBS for 24 h at 4  $^{\circ}$ C in a humid chamber. Similarly, C6 glioblastoma cells were incubated with anti-GFAP, anti-MAP2, and anti-mortalin (pan-cytoplasmic expression) antibodies. The cells were then exposed to secondary antibody for 2 h at room temperature. Anti-fading reagent was used to mount cells. The cell images were captured and analysed by employing Pro-Plus software.

#### 2.5.8. Autophagy assay by Alexa Fluor™ 488 cadaverine uptake

IMR-32 cells were seeded in T25 flasks, trypsinized and harvested by centrifugation after treatment with or without shankpushpi (2%) for 48 h. Next, the cell pellet was homogenized in 0.5 ml of homogenization buffer (250 mM sucrose, 1 mM EDTA, 10 mM HEPES) pH 7.0 along with protease inhibitors. The homogenate was centrifuged at 1000 $\times$ g for 5 min at 4  $^{\circ}$ C. The supernatant was collected and incubated for 10 min at 4  $^{\circ}$ C with Alexa Fluor™ 488 cadaverine (Catalog: A30676, Thermo Fisher Scientific) at a final concentration of 25  $\mu$ M. After incubation, the supernatant was centrifuged down at 20000 $\times$ g for 20 min at 4  $^{\circ}$ C. The pellet was then washed and resuspended in 600  $\mu$ L resuspension buffer (140 mM KCl, 10 mM MgCl<sub>2</sub>, 5 mM KH<sub>2</sub>PO<sub>4</sub>, 1 mM EGTA, 10 mM MOPS) and the 300  $\mu$ L sample was loaded on a 96-well black flat bottom plate. The sample fluorescence representative of cadaverine uptake was measured in a Tecan microplate reader M200 at 495 nm excitation and detected at 519 nm emission.

### 2.5.9. Western blot

IMR-32 cells were harvested post treatment with shankhpushpi, homogenized in the following buffer (50 mM Tris, 150 mM NaCl, 1 mM EDTA, 100  $\mu$ M sodium vanadate, 1 mM PMSF and 0.5 mM DTT), centrifuged and protein concentration was determined by the Bradford method. 30  $\mu$ g of protein was separated on 10% SDS-polyacrylamide gel electrophoresis (SDS-PAGE) and transferred to PVDF membrane (Hybond-P) through a semidry Novablot system (Amersham Pharmacia). Later, the blot was antibody probed with mouse anti-mortalin and anti-GFAP monoclonal antibodies for 2 h. These were imaged using an ECL Plus Western blot detection system (Amersham Biosciences).  $\beta$ -actin was used in each sample to account for variations across samples.

### 2.5.10. Statistical analysis

For statistical analysis purposes, GraphPad Prism (version 8.0.1) software was adopted to analyse and graph the results by Student's *t*-test to determine the significance of the means. Values are expressed as the mean  $\pm$  SEM or mean  $\pm$  SD, with  $p < 0.05$  considered statistically significant.

## 3. Results

### 3.1. Phytochemical profiling of shankhpushpi extracts

Phytochemical profiling of aqueous extract of shankhpushpi was performed in LC-MS. The profile revealed the presence of phyto-compounds such as Astragalin, Clitorin, Kaempferol and Kaempferol-3-O-robinobioside (Table 1, Fig. 1A and Supplementary file).

### 3.2. Effect of shankhpushpi on cytotoxicity and cell cycle in IMR32 neuroblastoma cells

The cytotoxic effects of the Shankhpushpi on cells were assessed to decipher non-toxic doses for the succeeding experiments. IMR32 neuroblastoma cells were treated with different concentrations (1–6%) of aqueous extract of Shankhpushpi (*Clitoria ternatea* L.) for 24, 48 and 72 h. The viability of cells was monitored by MTT assay, and the results showed that concentrations higher than 4% insignificantly reduced the viability of IMR-32 cells at 72 h (Fig. 1B). Higher concentrations of shankhpushpi (>2%) treatment of C6 rat glioblastoma cells for 4–5 days resulted in death, as indicated by changes in cell morphology (Fig. 1C). Furthermore, the flow cytometric analysis of cell cycle after 48 h clearly showed no significant differences with respect to the number of cells at different phases, indicating that shankhpushpi extract lacks its effect on cell proliferation (Fig. 1D). Hence, in the present experiments, the subtoxic doses of 1% or 2% shankhpushpi aqueous extract were used to study the effects on differentiation, DNA repair and other parameters in human IMR32 neuroblastoma cells and rat C6 glioblastoma cells, respectively.

### 3.3. Enhancement of in vitro neurodifferentiation by shankhpushpi

The aqueous extract of *Clitoria ternatea* L. induced the expression of neuronal differentiation markers, viz., GAP43, MAP2, NF-H and PSD markers, in IMR32 cells (Fig. 2A) and MAP2, GFAP and mortalin in C6 glioblastoma cells (Fig. 2B), indicating that there is no immediate

toxicity by shankhpushpi if treated for 4–5 days. Additionally, concentration dependent increase in the neuronal marker's expression was observed in IMR32 cells (Fig. 2A). The data thus showed enhanced cellular differentiation in human neuroblastoma and rat glioblastoma cells (Fig. 2) indicating neuroprotective role of Shankhpushpi.

### 3.4. Free radical scavenging activities of Shankhpushpi and analysis of its role in mitochondrial functions in glioblastoma cells

To assess the neuroprotective potential of shankhpushpi free radical scavenging assays such as DPPH and ABTS were performed. The free radical scavenging activity of aqueous extract of *Clitoria ternatea* L. was carried out using spectrophotometry. The extent of scavenging activities of DPPH and ABTS was quantified using different concentrations of aqueous extracts of shankhpushpi with L-ascorbic acid as the standard compound. The results indicated that the scavenging effect of the *Clitoria ternatea* L. extract increased with increasing concentration (Fig. 3). Flow cytometric analysis of mitochondrial functions showed that the treatment of IMR-32 cells with 2% shankhpushpi aqueous extract for 24 h elevated mitochondrial membrane potential and reduced reactive oxygen species. On the other hand, mitochondrial mass was marginally increased by aqueous extract of Shankhpushpi (Fig. 3). The mitochondrial functional assays indicate ROS clearance and neuroprotective ability of shankhpushpi in neuronal cells.

### 3.5. Influence of shankhpushpi on constitutive and induced DNA repair capacity in IMR32 cells

The DNA repair capacity of the cells treated with shankhpushpi was assessed using comet and cell-free base excision repair assays. The constitutive base excision repair capacity was found to be insignificant in shankhpushpi extract-treated cells when compared to control cells (Fig. 4A, B, 4C, 4D). The induced BER repair capacity was assessed by exposing the shankhpushpi-treated cells to 1J/cm<sup>2</sup> UV irradiation. Interestingly, the induced BER capacity increased significantly in a time-dependent manner (Fig. 4A, B, 4C, 4D). The results of olive tail moment analysis showed that there was significantly higher repair at 2 h in shankhpushpi-treated cells, indicating enhancement of UV induced DNA repair (Fig. 4E). Further phospho- $\gamma$ H2AX immunofluorescence staining and analysis by flow cytometry revealed no DNA DSBs in shankhpushpi treated IMR-32 cells (Fig. 4F and G). Thus, the overall data indicated enhancement of DNA repair capacity by shankhpushpi.

### 3.6. Effect of shankhpushpi on autophagy in IMR32 neuroblastoma cells

Cadaverine uptake was assessed to determine the ability of *Clitoria ternatea* L. plant on autophagy in neuroblastoma cells. The results of the analysis of autophagy by Alexa Fluor<sup>TM</sup> 488 cadaverine uptake clearly showed a significant reduction in the level of autophagy by shankhpushpi (2%) in IMR-32 cells at 24 h (Fig. 3F), indicating that this reduction increases the quality control of the cells.

## 4. Discussion

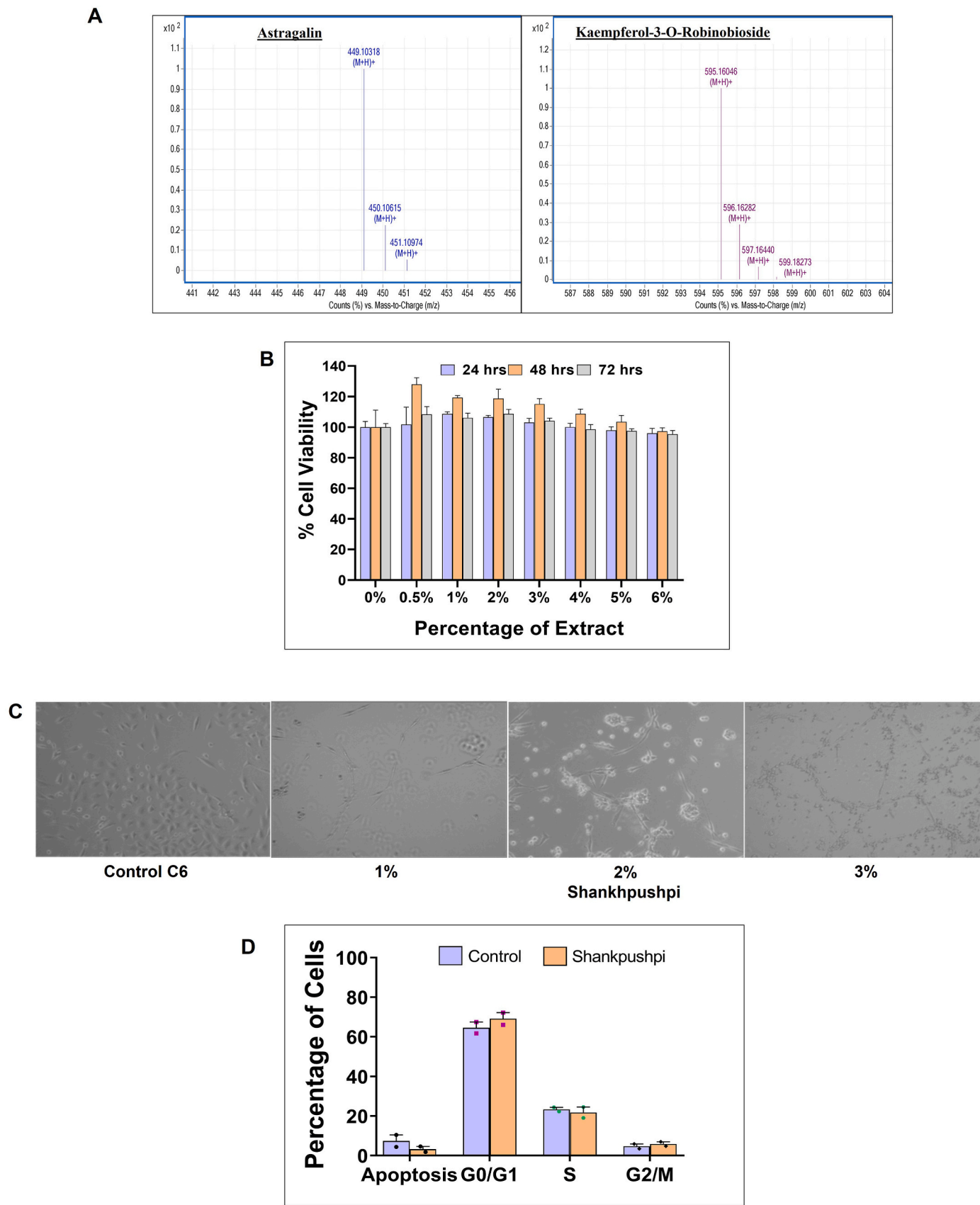
Traditional medicine in India is represented by the widely renowned system of Ayurveda, which comprises all aspects of human life that includes geriatrics, nutrition, rejuvenation and others. Rasayana therapy,

**Table 1**

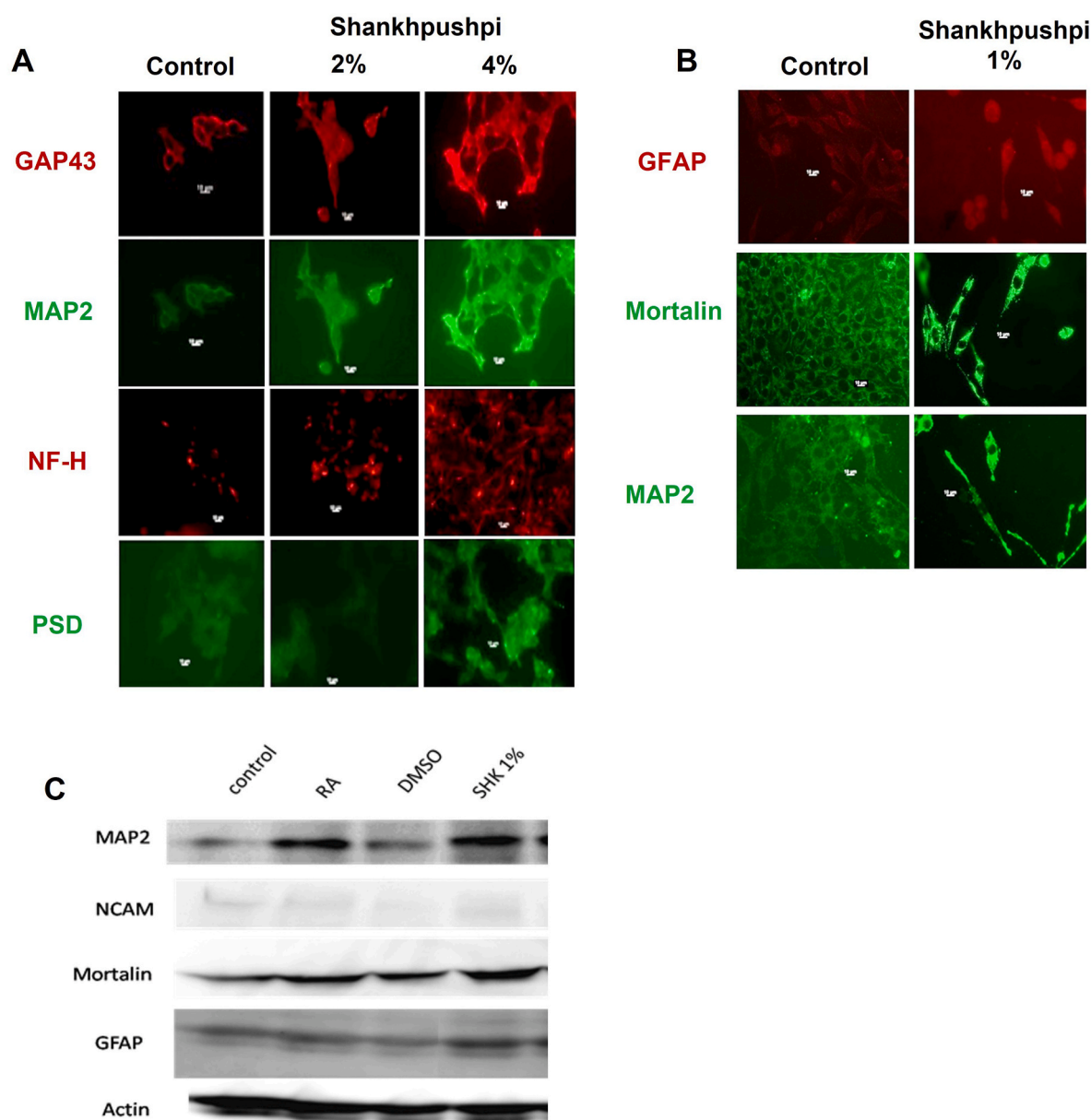
Phytochemical profile (identified phytochemicals) of Shankhpushpi aqueous extract performed in LC-MS.

| Phytochemicals in <i>Clitoria ternatea</i> L. | Reference      | PPM Error | Mass     | Avg Mass | m/z      | RT     | Vol % |
|-----------------------------------------------|----------------|-----------|----------|----------|----------|--------|-------|
| Kaempferol                                    | Metlin/Pubchem | 8.29      | 286.0453 | 286.2476 | 287.0526 | 24.257 | 0.31  |
| Astragalin                                    | Metlin/Pubchem | 6.70      | 448.0959 | 448.3576 | 449.1032 | 22.521 | 0.50  |
| Kaempferol-3-O-robinobioside                  | Metlin/Pubchem | 4.13      | 594.1532 | 594.5022 | 595.1605 | 22.576 | 3.58  |
| Clitorin                                      | Metlin/Pubchem | 0.08      | 740.2087 | 740.6353 | 741.2161 | 21.477 | 2.81  |





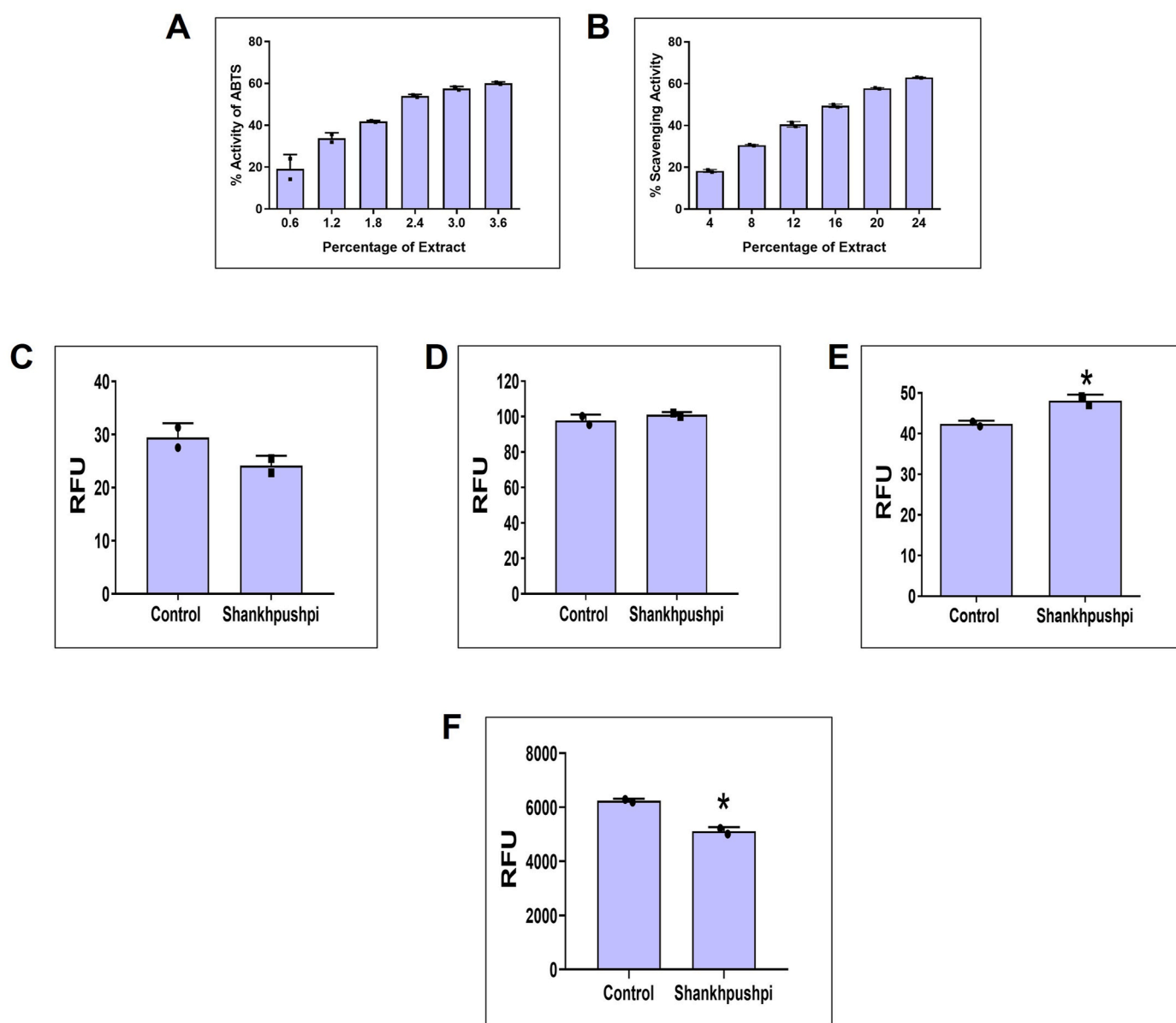
**Fig. 1.** A) Mass spectrogram for the selected phytochemicals found in Shankpushpi aqueous extracts. B) % Cell viability analysis by MTT assay in IMR-32 cells after treatment with Shankpushpi aqueous extract at 24 h, 48 h and 72 h. C) Bright field images of C6 rat glioblastoma cells after 5 days of the treatment regimen. D) Cell cycle analysis of IMR-32 cells.



**Fig. 2.** Immunofluorescence images of the expression of differentiation markers in **A)** IMR-32 cells and in **B)** C6 glioblastoma cells after treatment with Shankpushpi aqueous extract. **C)** Western blot images of the expression of differentiation markers in C6 glioblastoma (SHK – Shankpushpi aqueous extract).

which is one of the branches of Ayurveda, is said to fortify the physical, mental and spiritual aspects of one's life. Through the rejuvenation and renovation of the bodily tissues and the fortification of the nutritional status, *rasayanas* enhance qualities such as cognitive function, intellectual ability, longevity etc [2,33]. Essentially the therapy induced through *rasayana* tantra has its basis in molecular nutrients and nutrient enhancing agents which in turn act through three basic mechanisms such as *Rasa* enhancing (direct nutrient effect), *Agni* enhancing (promoters of digestion and metabolism), and *Srotas* purifying agents (promoters of microcirculation and tissue perfusion) [3]. There are traditional herbal formulations used individually or in combination with other plant species as *rasayana*. Medhya *rasayana*, known as nerve tonics, have been said to improve and fortify mental health. Shankpushpi is one such medhya *rasayana* plant. The nootropic effect of shankpushpi is reported at the clinical or preclinical level [8,27]. *Clitoria ternatea* L., one of the four designated shankpushpi plants, is used in the preparation of medhya *rasayana*/brain tonic and its use is reported to possess several useful health effects [8,12,14,17,18]. Even

though the therapeutic effect is known and practiced by Ayurvedic practitioners, the implicated molecular mechanisms are not completely understood. Understanding the molecular mechanisms/effects behind the medicinal value of traditional medicines is essential to enhance knowledge and help refine medical research and find novel strategies to manage various diseases and disorders. Hence, in the present studies, *Clitoria ternatea* L., was chosen from the others based on the proposed therapeutic effect, the geographical location of the experiments and the fact that there exists limited scientific literature enumerating the effects of this plant, relative to the others that form shankpushpi. In this study therefore, we investigated the *rasayana* like properties of *Clitoria ternatea* L. and assessed its molecular mode of action in the brain cells *in vitro*. To understand the mechanisms, an attempt has been made to identify the effects of aqueous extracts of shankpushpi on neuroblastoma cells *in vitro*. Although the use of shankpushpi prepared using *Clitoria ternatea* L. is found to be safe, studies have also reported its toxic effects [23,34]. Hence, we examined the cytotoxicity of shankpushpi aqueous extracts on neuroblastoma (IMR-32) cells. Our results indicated



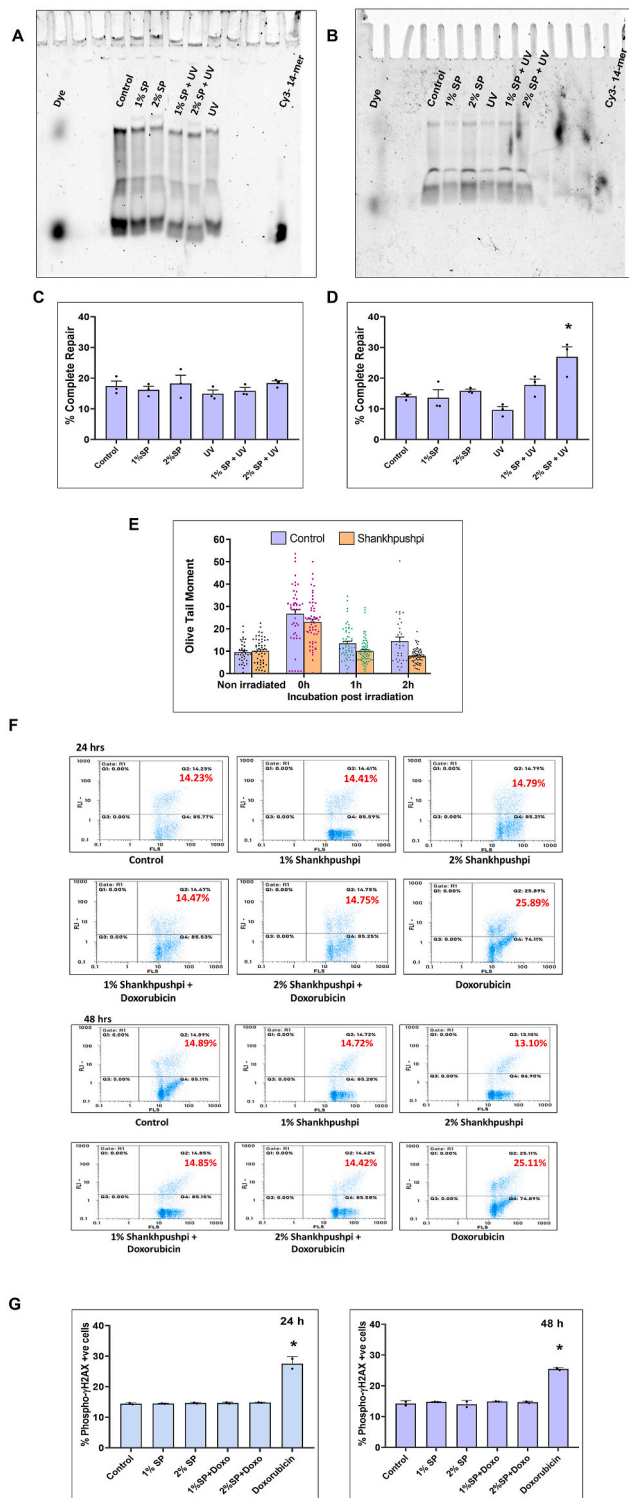
**Fig. 3.** Free radical scavenging activity **A)** ABTS and **B)** DPPH; mitochondrial function assay **C)** ROS, **D)** mitochondrial mass (MM), **E)** mitochondrial membrane potential (MMP) of shankhpushpi aqueous extract. **F)** Analysis of autophagy using Alexa Fluor cadaverine uptake in IMR-32 cells treated with or without shankhpushpi (2%) for 24 h. The data are represented as relative fluorescence units, and statistical significance is represented as asterisks \*  $p < 0.05$ . All values are expressed as the mean  $\pm$  SD; \*  $p < 0.05$ .

no significant cytotoxicity even at high concentrations of shankhpushpi aqueous extract (6%) at three time points tested on the neuroblastoma cell line (Fig. 1B). Decrease in the cell viability with increasing concentrations of *Clitoria ternatea* L. extracts is reported [34,35], which is attributed to the presence of alkaloids, tannins, saponins, etc., Based on the cytotoxicity results and the literature, nontoxic dose of shankhpushpi (2% aqueous extract), was used in our *in vitro* experiments.

Understanding the differentiation block in cancer cells and its modulation holds promise in cancer therapy. Therefore, the differentiation of neuronal tumor cells with shankhpushpi extracts into neuronal phenotypes was studied. Our results indicated the induction of differentiation in C6 rat glioblastoma cells by 2% shankhpushpi extract (Fig. 1D). Furthermore, analysis of markers of differentiation illustrated the expression of GAP43, MAP2, NF-H and PSD proteins in IMR 32 cells and GFAP, MAP2, and Mortalin (pan-cytoplasmic expression), a chaperone protein in C6 glioblastoma cells by *Clitoria ternatea* L. aqueous extracts (Fig. 2). Much evidence is available on the increased expression

of these proteins in differentiated neuronal cells along with their role in the maturation of neurons, cell proliferation, migration, axonal growth, morphogenesis, synaptic plasticity and tumor growth suppression [36–42]. This gives a plausible role of differentiation in cancer therapy. Studies have also reported miRNAs such as miR-34, miR-200 and miR-221/222 with a putative role in differentiation [43]. Studies are required to elucidate the target(s) of action of shankhpushpi rasayana and its phytoconstituents to explain the differentiation potential.

It is well known that the imbalance between ROS and antioxidant defense mechanisms results in oxidative stress which in turn leads to damage to almost all macromolecules, including DNA [44]. The persistence of these lesions leads to apoptosis and, in the case of neurons, neurodegeneration [45], which is one of the hallmarks of brain ageing [46]. Hence, the analysis of antioxidant activity of shankhpushpi is indispensable, and the enhanced antioxidant activity of shankhpushpi extract could be the key to its healing abilities. In the present investigations, we observed an increased free radical scavenging activity



**Fig. 4.** DNA (BER) repair assay in IMR-32 cells after treatment with (1%, 2%) Shankpushpi aqueous extract, (1J) UV and (1%, 2%) Shankpushpi aqueous extract + (1J) UV. Representative Urea PAGE gel images of BER assay for A) 24 h B) 48 h; Graph - Percentage complete base excision repair at C) 24 h and D) 48 h. Modulation of UV (1J/cm<sup>2</sup>) induced DNA damage/repair was observed in IMR-32 cells by Shankpushpi extract analysed by E) Comet assay. F) Phospho-γ-H2ax immunostaining, Flow cytogram, G) graph showing no significant change in the treatment groups (1%, 2% Shankpushpi aqueous extract and 1%, 2% Shankpushpi aqueous extract + 50 nM doxorubicin) after 24 h and 48 h (SP - Shankpushpi aqueous extract, Doxo - doxorubicin). All values are expressed as the mean ± SEM, \*p < 0.05.

of shankpushpi aqueous extract in a concentration-dependent manner (Fig. 3A and B) on par with many reports explaining the strong antioxidant potential of *Clitoria ternatea* L. [23,34,35,47,48]. and other medhya plants [49–52]. The antioxidant activity/potential of medhya rasayana plant *Clitoria ternatea* L. is largely attributable to the action of individual phytochemical constituents such as Clitorin, Kaempferol or others [53], and the mechanism of action of such phytoconstituents remains to be fully explored.

Mitochondria are essential organelles that control a wide range of functions [54,55]. It is known that the loss of mitochondrial membrane potential ( $\Delta\Psi_m$ ), which is critical for the physiological functions of mitochondria, results in the depletion of energy in cells, which causes cell death [56]. There are a substantial number of studies that have firmly documented the relationship between ROS and mitochondrial membrane potential [56,57]. The decrease in mitochondrial membrane potential is associated with increased production of ROS, as observed in Fanconi anemia cells [57]. The maintenance of mitochondrial membrane potential with sufficient ROS is essential for cell proliferation. Recent evidence further indicates that the biogenesis of mitochondria and their regulation are important factors in the survival and recovery of neuronal cells in neurodegenerative disorders [58,59]. Thus, we assessed the role of shankpushpi (as it possesses excellent free radical scavenging activity) on ROS generation, mitochondrial membrane potential and mitochondrial mass in neuroblastoma cells, which are implicated in the quality of the cells. Our results demonstrated reduction in reactive oxygen species (Fig. 3C) associated with increased mitochondrial potential (Fig. 3D) and a marginal increase in mitochondrial mass (Fig. 3E). The reduced ROS by shankpushpi extract may be attributed to its various phytochemicals showing excellent free radical scavenging activity [53]. ROS formation generally occurs at complex I and complex III, which depends on proton motive force, NADH:NAD ratio and contributions from mitochondrial membrane potential [60]. Chinopoulos et al. (2001) showed that the inhibition of complex I stimulated ROS generation, which consequently collapsed the mitochondrial membrane potential [61]. Mitochondrial membrane potential is also implicated in mitochondrial homeostasis and the removal of dysfunctional mitochondria. It is responsible for the transport of ions and proteins essential for the proper functioning of mitochondria and plays an essential role in multifarious cellular processes, including DNA repair, DNA replication, protein synthesis, electron transport, and cofactor biosynthesis [62]. Increased mitochondrial ROS also aids DNA DSB generation in telomeres [63]. Furthermore, a higher MMP results in a higher energy capacity of the inner mitochondrial membrane, and ATP synthesis and mitochondrial proteins are involved in MMP [64]. The increase in MMP by shankpushpi extract may be due to a reduction in ROS and an increase in mitochondrial proteins, which needs to be tested. Consistent with our results, Xu et al. (2012) demonstrated a reduction in ROS generation and restoration of mitochondrial membrane potential by pretreatment with asiatic acid from *Centella asiatica* (L.) Urb. (another medhya plant) in SH-SY5Y neuroblastoma cells [65]. Yu et al. (2014) also showed that quercetin, another compound present in various plants, including *Clitoria ternatea* L. has been found to reduce H<sub>2</sub>O<sub>2</sub>-induced ROS by sustaining antioxidant enzymes with elevated mitochondrial membrane potential, thereby maintaining mitochondrial functions in mouse embryos [66]. Several studies in the recent past using plant extracts have delineated the importance of ROS and mitochondrial membrane potential in cell survival or death [67–70].

Exposure to a vast array of either endogenous or exogenous DNA damaging agents results in multiple types of DNA damage in cells. However, the cells possess inbuilt DNA repair machinery to combat the effects of these agents. The functional quality of cells depends on the integrity of genetic material. Genomic instability plays a significant role in aiding ageing and age-related diseases. Accumulated DNA damage induces multiple age-related effects, such as cell death, senescence, tissue dysfunction, and inflammation [71]. Premature ageing, wherein the subjects exhibit, many physiological ageing features are caused mainly



due to failure of the DNA damage response (DDR) and repair mechanisms such as DNA double strand break repair (DSBR), nucleotide excision repair (NER), homologous recombination (HR) and BER [63]. Deficiencies of these DNA repair mechanisms cause early greying of hair, skin atrophy and others. Some examples of age-related diseases due to the collapse of DDR are ataxia-telangiectasia caused by DSBR deficiency, xeroderma pigmentosa due to NER deficiency and Werner syndrome caused by HR/BER deficiencies [63]. Further evidence suggests that epigenetic modulation and epigenetic memory loss through repeated DNA damage also induce ageing [72]. For instance, DNA methyltransferase (DNMT1) is present in DNA repair sites. DNA methylation patterns are perturbed with persistent HR and BER due to repeated DNA damage [72]. Furthermore, the poly ADP-ribosylation process that completes the BER also decreases cellular levels of  $\text{NAD}^+$ , leading to cell ageing or cell death by activated autophagy [72,73]. To overcome these age-related deficiencies, Ayurvedic rasayana formulations that show tissue rejuvenating properties, viz. medhya rasayana plant shankhpushpi were evaluated. Subsequently, we investigated the BER-enhancing properties of shankhpushpi extracts. Our results show a consistent increase in constitutive and induced base excision repair upon shankhpushpi extract treatment in IMR 32 cells in a time-dependent manner ( $p < 0.05$ ) without an increase in autophagy (Fig. 4A, B, 4C, 4D, 3F). This may overcome some of the age-related molecular signatures due to BER deficiency that warrant further mechanistic investigation. At this stage, we hypothesize that *Clitoria ternatea* L. due to its rasayana like properties may enhance DNA repair effectively through BER, thereby playing a role in the viability and functional quality of cells. The pretreatment of IMR 32 cells with shankhpushpi and then exposure to UVC radiation also showed a significant decline in DNA breaks as analysed by comet assay ( $p < 0.05$ ) (Fig. 4E), as well as no increase in phospho- $\gamma$ -H2AX immunostaining (Fig. 4F and G), indicating enhancement of DNA strand break repair by shankhpushpi in IMR-32 cells. These findings corroborate the enhancement of induced DNA repair capacity in IMR-32 cells by shankhpushpi. This is the first study of shankhpushpi (*Clitoria ternatea* L.) on IMR-32 cells showing enhancement of DNA repair through BER. Concurrent reports are available on the reduction of DNA damage by other medhya rasayana formulations [9,74–76].

Autophagy is a conserved dynamic catabolic process wherein functionally defective macromolecules and cell organelles are degraded, engulfed or recycled, thus helping to maintain cellular homeostasis [77, 78]. Recent evidence indicates that autophagy is induced in response to DNA damage, increased ROS and alteration of gene expression, thereby playing a role in cellular protection, homeostasis and longevity or cell death [78]. Furthermore, it is implicated in neurodegeneration, diabetes, immunological disorders, cancer and ageing, thereby illustrating its importance in the quality and health of cells. Thus, it is imperative to understand the effects of shankhpushpi on autophagy, as shankhpushpi is postulated to improve brain cell functions. The data in the present study clearly illustrated the reduction of autophagy by shankhpushpi extract in IMR-32 cells compared to the respective untreated cells (Fig. 3F), thus offering its protective role. On par with these findings, Yan et al. (2016) explained that luteolin (LUT), a flavone present in almost all Chinese herbs, significantly reduced UV A-induced autophagy and the expression of autophagy-related genes such as *HIF-1 $\alpha$* , *LC3-II* and *Becn1* in human skin fibroblasts via a decrease in ROS [79]. Many reports are available on the association of low levels of autophagy with cellular survival. Chen et al. (2016) showed inhibition of autophagy in A549 cells by inhibitors such as 3-methyl adenine and spautin-1 or by knocking out *ATG5* and *BECN1* using siRNAs, which in turn decreased the hypoxia-induced cell death [80]. Hypoxia and consequent oxygenation of H9c2 cells resulted in increased autophagy and apoptosis, whereas downregulation of miR-21 microRNA and the overexpression of miR-21 enhanced cell survival by reducing autophagy [81]. Wang et al. (2014) listed Chinese medicines with phytoconstituents that induce autophagic cell death [82]. These data clearly indicate a dual role of

autophagy, where basal or low levels of autophagy have an impact on cell survival by maintaining normal functions and homeostasis. On the other hand, elevated autophagy results in cell death [83,84]. There are many reports that suggest that under extreme stress conditions, high levels of autophagy induce autosis or autophagy-dependent cell death [85]. One report by Zhang and coworkers (2011) showed that rapamycin treatment increased motor neuron degeneration in mouse models with amyotrophic lateral sclerosis [86]. Increased autophagy is reported in acute CNS diseases such as brain hypoxia ischemia, brain stroke and brain injury [87]. Our results show a reduction in both autophagy and cellular ROS with increased BER activity, which thus indicates enhancement of DNA repair and potential delay in age-related dysfunctions at least at the autophagy and ROS levels (Fig. 3F, C, 4A, 4B, 4C, 4D). One plausible explanation for these events is that succinct evidence suggests that increased ROS generation induces autophagy [88]. In contrast, in our study, there was decreased ROS generation, which may have possibly helped in the cessation of autophagy in shankhpushpi treated cells. Hence, we hypothesize that the failure to repair the repeated DNA damage through DDRs over time as the cell ages, along with increased ROS generation subsequently leads to either excessive autophagy leading to cell death or senescence and other age-related dysfunctions, whereas shankhpushpi helps reverse this process. Furthermore, increasing evidence from preclinical studies suggest that autophagy modulators can be potential suppressors of age-related neurodegenerative diseases [89].

## 5. Conclusion

The quality and viability of cells depends on the proper functioning of the cellular machinery. DNA repair machinery, autophagy, antioxidant defence, and cell proliferation are among the many others implicated on functional aspects of cells. There are reports to show the intricate relationships among several pathways, including ROS, DNA repair, autophagy, cell survival, neurogenesis, stress, and regulation of these by multiple genes. All these factors have an impact on ageing and longevity. The overall results of the present *in vitro* experiments clearly demonstrated the alleviation of autophagy and reactive oxygen species, increase in DNA repair and mitochondrial membrane potential and insignificant changes in mitochondrial mass by shankhpushpi (*Clitoria ternatea* L.). Additionally, shankhpushpi also showed the differentiation potency of neuroblastoma cells. The present data highlights the protective role of shankhpushpi by refining and improving the functional qualities of neuronal cells. Thus, establishing its rasayana like properties and resolving its molecular mechanisms.

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## Data availability

All data generated or analysed in this study are included in this article.

## Ethical approval

Not applicable.

## Author contributions

KP: Investigation, Data collection, Visualization, Data analysis, Writing-review and editing draft. KSR: Investigation. SM: Writing-reviewing draft. RW: Neuronal marker analysis, Supervision,

Investigation. SCK: Supervision, Investigation. KS: Conceptualization, Methodology, Supervision, Funding. GKP: Conceptualization, Methodology, Supervision, Funding, Writing – original draft.

## Declaration of generative AI in scientific writing

None.

## Conflict of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix ASupplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jaim.2024.101050>.

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